

SatQuery — Talk to the Earth. Understand the Change.

AI-assisted geospatial console for satellite imagery. Plain language → map highlights + answers.

1. Problem & Vision

Satellite archives (Sentinel-2, 10 m) are rich but require GIS expertise. **SatQuery** lets anyone *talk* to the Earth: ask “what’s visible here?”, draw a region, scrub time, and get instant land-cover answers — no GIS syntax.

2. Five Core Features

#	User Action	What Happens
1	*“What’s visible here?” / “Find water bodies”*	Detect & highlight — vectorized polygons (water / vegetation / built-up) overlay the map + text reply
2	Timeline 2018 → 2026 slider	Swaps NASA GIBS VIIRS imagery; AI narrates change (e.g. <i>*vegetation -27%, built-up +41%*</i>)
3	Draw polygon (or 3-point triangle)	Region summary — water / vegetation / built-up % + changed? flag via GIBS RGB classification
4	*“Areas near the river where construction increased”*	Plain-language spatial query — LLM parses intent (detect/quantify/change/overlay/list) → spatial ops (buffer, intersection)
5	Proactive alerts	Flags unusual recent change; jump to before/after

Bonus: **Split-screen swipe (2021 vs 2026)** — draggable curtain (clip: rect) compares two dates; **One-click PDF** — dark aerospace A4 report.

3. Live Demo Flow (60 s)

- 1
- Landing — 3D Earth (EarthScene3D + LiveCoordinates) → **Enter workspace**
- 2
- Search** Chennai (Nominatim) or **Triangle** Lat1 Lon1 ... → map flies, draws polygon
- 3
- Chat** “Find water bodies” → highlights Bay of Bengal
- 4
- Draw region** → Evidence panel shows bars + CHANGE 2021→2026
- 5
- Timeline** scrub + **Split** toggle → curtain compare
- 6
- Export** ■ EXPORT EVIDENCE REPORT → SatQuery_Intelligence_Report_2026-08-31.pdf

4. Tech Stack

Layer	Choice
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Frontend	Next.js 16 (App Router, React 19, TS), Tailwind 4, Zustand, Framer Motion
Map	Leaflet + react-leaflet, NASA GIBS WMTS (VIIRS TrueColor, BlueMarble)
Backend	FastAPI + Uvicorn, Rasterio/Shapely, Pillow, NumPy
AI	Claude 3.5 Haiku → Gemini 3.6 Flash fallback, strict guardrail + 70-case harness
Export	jspdf + html2canvas-pro (lab-color safe)
Data	GIBS live tiles + synthetic 50×50 GeoTIFFs for fallback

5. Architecture (one glance)

```
Next.js ( :3000 ) → FastAPI ( :8000 ) → GIBS / Dynamic World
MapStage/MapView ↔ /api/query, /detect, /timeline, /analyze, /geocode, /places
ChatPanel/Timeline ↔ llm_service (Claude→Gemini) + gibs_analyzer (RGB → classes)
EvidencePanel      ↔ regionResult {water,vegetation,built_up, change}
```

State: lib/store.ts (geometry, highlights, splitEnabled/position, regionResult). One Leaflet map — split via clip: rect, never re-inits.

6. What's Built & Validated

- **70 validated test cases** (data/validated_test_cases_70.csv via known_cases.py — e.g. *Chennai water 265 km²* ±5%)
- **Cloud mask** (brightness>0.88), **high/low veg split**, **parallel tile fetch** (~2 s)
- **CORS-safe PDF** with map snapshot, metrics bars, AI narrative
- **Mobile + desktop** (pointer events, flyToBounds, no memory leaks)

7. Run Locally

```
# Backend
cd satquery && .venv/bin/uvicorn main:app --port 8000 # needs GEMINI_API_KEY
```

```
# Frontend
cd "satquery 2" && npm run dev # http://localhost:3000 → /workspace
```

8. Known Limits & Next

- **No NIR** → built-up + bare combined; cloud mask simple; VIIRS ~375 m (coarse for tiny polygons)
- **Next:** 10 m Dynamic World tiles, auth/rate-limit, vector tile caching, multi-region compare

Tagline: *Talk to the earth. Understand the change.* — Vercel-minimal, void #0A0C0E + signal #8CFFBE.

Prepared for DevJams / presentation — 5 min pitch.